

Noor Khan

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Skoltech University, Moscow, Russia

Visiting Researcher, KFUPM, Dhahran, Saudi Arabia | Nationality: Pakistan

Research Profile

MSc candidate in Energy Systems at Skoltech (graduating June 2026), specialising in stability and adaptive control for low-inertia power systems. Skilled in RTDS, MATLAB/Simulink, PSCAD, and hardware-in-the-loop (dSPACE). First author of one published IEEE paper and two under review: IEEE T-PWRS and ICRERA 2026.

Education

MSc Energy Systems Sep 2024 – June 2026
Skolkovo Institute of Science and Technology (Skoltech), Moscow, Russia
GPA: **3.90/4.00**
Advisor: Prof. [Oleg Khamisov](#)

BE Electrical Engineering (Gold Medalist) Jan 2019 – Dec 2022
Sukkur IBA University, Sukkur, Pakistan

Publications (All First-Author Contributions)

[1] **Khan, N.** et al., An Optimal Contingency-Sensitive Inertia and Damping Control for Grid-Forming Inverters. *2025 7th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency (SUMMA)* Lipetsk, Russian Federation, 2025, pp. 906-911 DOI: [10.1109/SUMMA68668.2025.11302304](https://doi.org/10.1109/SUMMA68668.2025.11302304)

[2] **Khan, N.** et al., Optimal Adaptive Droop Control with AI-Based Contingency Detection for Frequency Regulation. Under Review, **IEEE Transactions on Power Systems (Q1, IF: 7.2)**. March 2026 [\[link\]](#)

[3] An Optimal Adaptive Control for VSM-Based Grid-Forming Inverters
Submitted to **ICRERA 2026, France** April 2026 [\[link\]](#)

Work Experience

Visiting Research Intern – Academic Mobility Program (Certificate) Jan 2026 – Present
[King Fahd University of Petroleum & Minerals \(KFUPM, QS Rank 67\)](#), Dhahran, Saudi Arabia

- Performed hardware-in-the-loop validation of advanced control for hybrid grids using dSPACE MicroLabBox.
- Conducted experimental analysis and worked toward the preparation of a journal publication.

Power System Protection & Automation Intern Jun 2025 – Jul 2025
[Tekvel, Moscow, Russia](#)

- Developed an IEC 61850-based virtual relay testing workflow using Sampled Values and GOOSE messaging.
- Awarded *Best Industrial Immersion Project 2025*.

Engineering Intern May 2023 – Jun 2023
[Power and Water Division \(PWD\), Skardu, Pakistan](#)

- Assisted in monitoring and performance assessment of hydropower generation units.
- Conducted fault analysis and contributed to upgrade recommendations and technical documentation.

Honours and Awards

- **Gold Medalist**, BE Electrical Engineering Sukkur IBA University, 2023
Graduated top-ranked in the cohort for academic excellence.

- **Winner – Best Industrial Immersion Project** Skoltech, 2025
Recognized for developing a real-time relay testing framework using Tekvel Magic.
- **Academic Mobility Scholarship** Skoltech, 2025
Awarded to one student for a funded research internship at a top-ranked university.
- **National Talent Hunt Program (NTHP) Scholarship** Sukkur IBA University, 2019
Fully funded merit-based scholarship awarded to top 100 students nationwide.
- **Third Place – National Book Review Competition** Sukkur IBA University, 2019

Projects

Digital Twin-Inspired AI-Integrated Adaptive Control for Hybrid Grids *Skoltech/KFUPM, 2025–26*

Developed a Digital Twin-inspired optimal control framework combining offline scenario-based optimization with real-time AI contingency classification for a hybrid grid. Under review, IEEE Transactions on Power Systems.

HIL Validation of Adaptive VSM-Based GFMI Control for MicroGrids *Skoltech/KFUPM, 2025–26*

Implemented a virtual governor-based adaptive droop strategy for VSM emulating synchronous generator dynamics and validated in hardware-in-the-loop (HIL) using dSPACE MicroLabBox II. Manuscript in preparation.

High-Fidelity EMT Modelling and Control of a GFM Inverter in RTDS *Skoltech, 2025*

Developed a high-fidelity EMT model of a grid-forming inverter in RTDS using a 2–5 μ s substep environment and implemented advanced VSM control strategies such as droop control, adaptive inertia, and damping control.

Dynamic Modelling of IEEE 9-Bus and 39-Bus Power Systems *Skoltech, 2024–25*

Developed nonlinear dynamic models in Python and validated against MATLAB/Simulink and RTDS results for optimisation-based inertia and damping control studies.

Conference Presentations

- **7th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency (SUMMA 2025)**, Lipetsk, Russia.
Presentation: “Optimal Contingency-Sensitive Inertia and Damping Control for Grid-Forming Inverters.”
- **IEEE Student Conference on Engineering, Science and Technology**, Sukkur IBA University, 2021.
Presented undergraduate project on hybrid energy system design and integration.

Technical Skills

Simulation & Real-Time	RTDS, dSPACE MicroLabBox II, MATLAB/Simulink, PSCAD
Power System Software	PSS®E, ETAP, LabVIEW
Programming	Python (NumPy, SciPy, Scikit-Learn, PandaPower, Matplotlib), C++
Design Tools	EAGLE (PCB), SolidWorks, AutoCAD

Volunteering & Leadership

International Student Representative , Skoltech	Sep 2024 – Sep 2025
General Secretary, IEEE Student Branch , Sukkur IBA University	Jan 2021 – Jan 2022
Executive Member, Sports & Adventure Society , Sukkur IBA University	Nov 2021 – Jul 2022
Content Writer , Read Pakistan	Aug 2020 – Feb 2021

Languages

1. English (Fluent) 2. Urdu (Fluent) 3. Burushaski (Native) 4. Dawoodi (Native) 5. Shina (Fluent)

For recommendations please contact my MSc advisor, Dr. Oleg Khamisov (O.Khamisov@skoltech.ru), my mentor, Dr. Andrey Churkin (a.churkin@imperial.ac.uk), and my KFUPM co-supervisor, Dr. Muhammad Majid Gulzar (muhammad.gulzar@kfupm.edu.sa).